Flexible BLE Pressure Sensing System

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Electrical Engineering 491B

Table of Contents:

Abstract - **This project presents the design, implementation, and validation of a flexible, multi-sensor embedded system for measuring tongue pressure distribution in dysphagia analysis. The system integrates a custom 2-layer flexible PCB, force-sensitive resistors (FSRs), and a Nordic nRF52840 microcontroller to acquire and transmit analog pressure data via Bluetooth Low Energy (BLE) for real-time monitoring. A modular architecture enables the sensing system to interface with a detachable programming module through a flexible printed circuit (FPC) connection.**

A key focus of this work was improving system reliability, robustness, and usability over a prior prototype. This was achieved through the redesign of the programming architecture, implementation of regulated dual-source power management, and systematic hardware debugging. Key challenges, including signal noise and unstable sensor readings, were identified and resolved through oscilloscope-based analysis, optimized PCB routing, and improved signal conditioning. The final system demonstrates improved signal stability, reliable wireless communication, and a compact, flexible form factor suitable for real-world applications.

I. INTRODUCTION

Dysphagia is a condition that impairs a person's ability to swallow, often due to irregular or insufficient tongue movement during the oral phase of swallowing. Accurate diagnosis requires understanding how pressure is distributed across the upper palate; however, existing measurement approaches are limited in spatial resolution and often rely on single-sensor systems that fail to capture detailed pressure patterns.

To address this limitation, this project focuses on the development of a flexible, multi-sensor embedded system capable of capturing real-time tongue pressure distribution across multiple points of contact. The system is designed to be compact, minimally intrusive, and capable of wirelessly transmitting data for external analysis, enabling a more detailed and accessible approach to dysphagia assessment.

From an engineering perspective, the primary challenge was not only achieving functionality within strict size and flexibility constraints, but also ensuring reliable operation in a high-noise, high-impedance sensing environment. This required careful consideration of sensor interfacing, power integrity, signal routing, and wireless communication stability. The resulting design reflects a system-level approach that prioritizes robustness, reusability, and reliable data acquisition under real-world conditions. A conceptual layout of sensor placement within the oral cavity is shown in Figure 1.

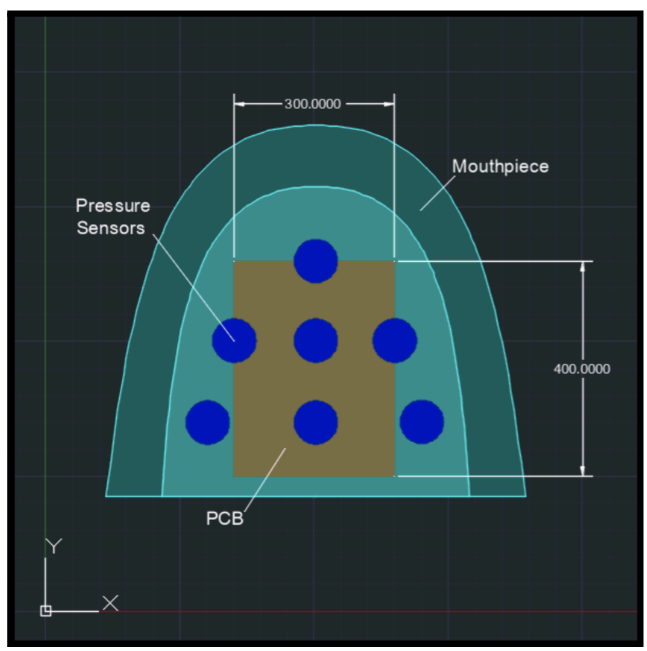

Figure 1. AutoCAD drawing of ideal product placement in mouth.

II. System Architecture & Engineering Design

The overall system architecture, including the programming interface and sensing subsystem, is shown in Figure 2.

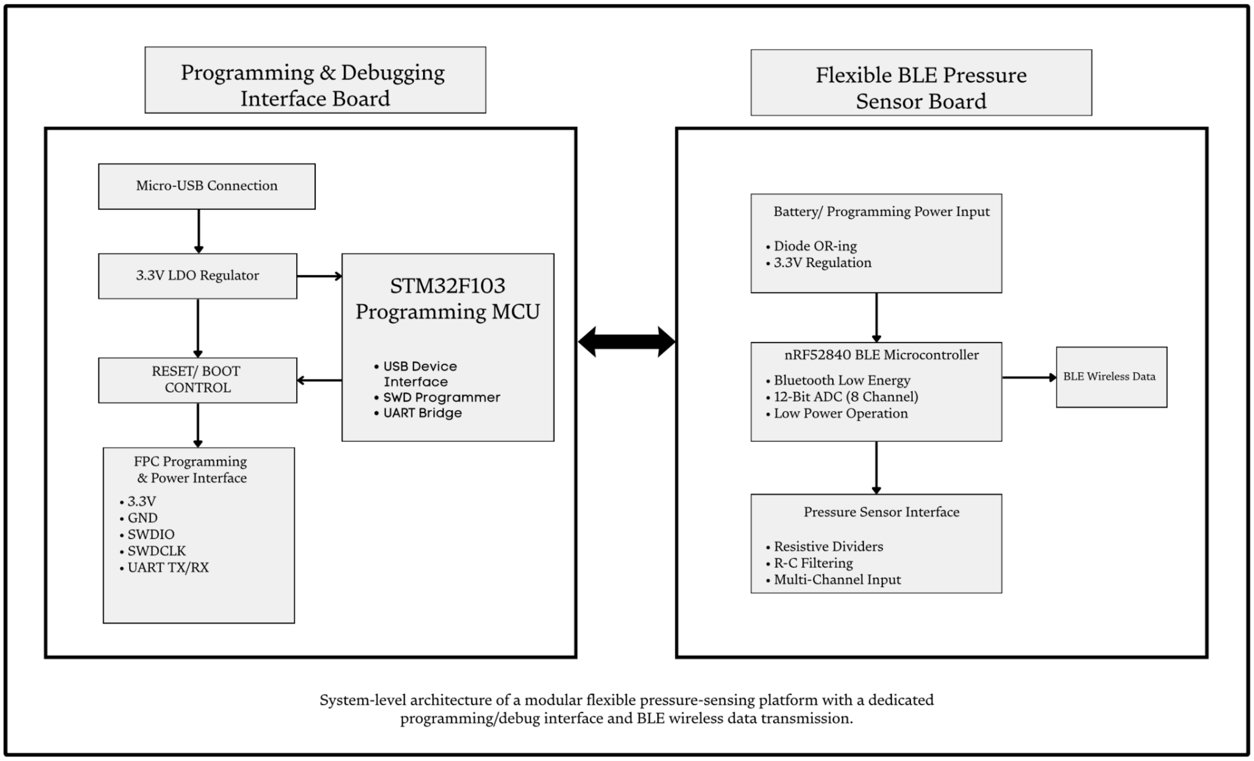

Figure 2. Flexible sensory system block diagram.

Programming & Debugging Interface (STM32-Based): The programming interface board is built around the STM32F103 microcontroller and serves as both a firmware programming tool and a debugging interface.

Key features include:

- Micro-USB interface for power and communication
- 3.3 V LDO regulation for stable operation
- SWD interface (SWDIO/SWDCLK) for direct programming
- UART bridge for communication and debugging
- Reset and boot control circuitry for firmware flashing

The programming interface board connects to the sensing system through a flexible printed circuit (FPC) interface, enabling power delivery and bidirectional communication. A key design improvement was the elimination of a one-time-use programming module, enabling repeated firmware updates and in-system debugging while significantly improving development efficiency and long-term system maintainability. This modular approach also improves fault isolation, allowing each subsystem to be independently tested and validated during development. Design emphasis was placed on maintaining signal integrity and stable wireless communications despite compact layout constraints and noise-sensitive analog inputs.

Flexible BLE Pressure Sensing Board: The sensing board integrates the nRF52840 microcontroller, selected for its BLE capability, low power operation, and integrated multi-channel ADC.
Key components include:

- nRF52840 BLE microcontroller
    - Bluetooth Low Energy communication
    - 12-bit ADC for multi-channel sensor input
    - Low power operation
- Pressure sensor interface
    - Resistive voltage divider networks
    - Multi-channel analog input
    - Optional RC filtering for noise reduction
- Wireless transmission
    - Real-time BLE data output

The design prioritizes compactness and flexibility to meet the necessary constraints while maintaining reliable analog signal acquisition and stable wireless communication under constrained conditions. Particular attention was given to minimizing noise coupling and maintaining signal integrity due to the high-impedance nature of the sensor interface. A representative reference design that informed the system form factor is shown in Figure 3.

Power Architecture & Protection: The system supports dual power sources: an external 3.7V lithium battery and programming interface power via USB.
To ensure safe and reliable operation:

- 3.3 V regulation is applied across both power paths
- Schottky diode OR-ing is implemented to prevent reverse current flow
- The system can safely operate with either or both power sources connected

This design protects the system during both development and deployment, preventing damage from improper power conditions while ensuring consistent and reliable operation across all use cases.

This approach also reduces the risk of unintended failure modes during debugging and field operation. The assembled hardware prototype with integrated pressure sensors is shown in Figure 4.

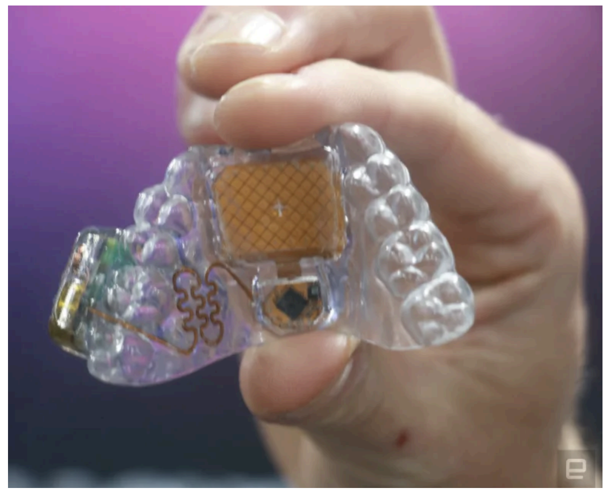

Figure 3. Engadget inspiration for final product

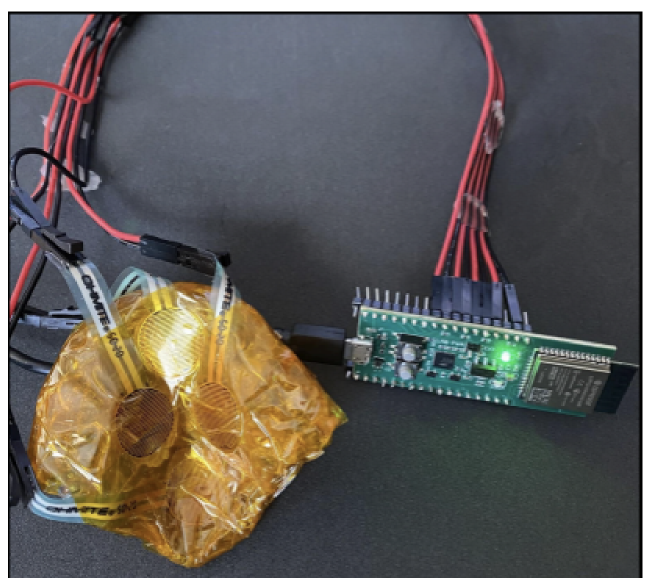

Figure 4. Product prototype with integrated force sensors.

III. Product Reliability & Debugging Analysis:

During development, several performance limitations were identified that impacted signal quality, measurement accuracy, and overall system reliability. These issues were not immediately apparent during initial design and required systematic debugging and validation to diagnose and resolve.

Signal Instability Observation: Initial testing of the sensor interface revealed unstable analog output behavior. Sensor readings fluctuated even under approximately constant applied force,with resistance exceeding 1 MΩ at low force levels. When used in a voltage-divider configuration, this results in high-impedance susceptible to noise and interference, the presence of noise and signal integrity issues. This inability was critical, as the system depends on accurate pressure measurements.

Root Cause Analysis: The instability was primarily attributed to the high-impedance nature of the sensor interface. The FlexiForce A201 force sensor operates as a variable resistance device, with resistance exceeding 1 MΩ at low force levels. When used in a voltage-divider configuration, this results in a high-impedance output node that is highly susceptible to noise and interference.

In this configuration, the output voltage became sensitive to:
- External electromagnetic interference
- Parasitic capacitance within the circuit and interconnects
- Coupling from nearby digital signals
- Noise introduced by an unfiltered analog signal path

These effects led to inconsistent voltage readings and reduced measurement reliability under steady input conditions.
Design Improvements: To address these issues, a low-pass filtering approach was implemented at the sensor output. A capacitor was added between the output node and ground, forming an RC filter with the existing resistor network. This modification attenuates high-frequency noise components prior to ADC sampling, improving signal stability.

Additional design considerations included:
- Improved signal routing to minimize analog–digital coupling
- Maintaining stable and regulated power delivery
- Enhancing overall signal integrity at the sensor interface

Validation Approach: A representative bench test was constructed to evaluate the impact of filtering on the high-impedance sensor interface. The setup utilized a FlexiForce A201 sensor in a voltage-divider configuration with a 1M Ω resistor and a 3.3 V supply, replicating the electrical behavior of the implemented design. Evaluation of the unfiltered configuration showed noticeable output fluctuation under approximately constant applied force. After introducing the capacitor between the output node and ground, the sensor output exhibited improved stability and a more consistent response. Following this, the resistor value was refined in order to achieve the clearest signal obtainable. Through analytical estimation and empirical validation, the resistor value was optimized to approximately 500 KΩ to achieve improved signal clarity while maintaining adequate sensitivity. This reduction in resistance lowered the output impedance, improving noise immunity while preserving sufficient sensitivity for accurate measurement. A representative comparison of sensor output stability under different interface conditions is shown in Figure 5.

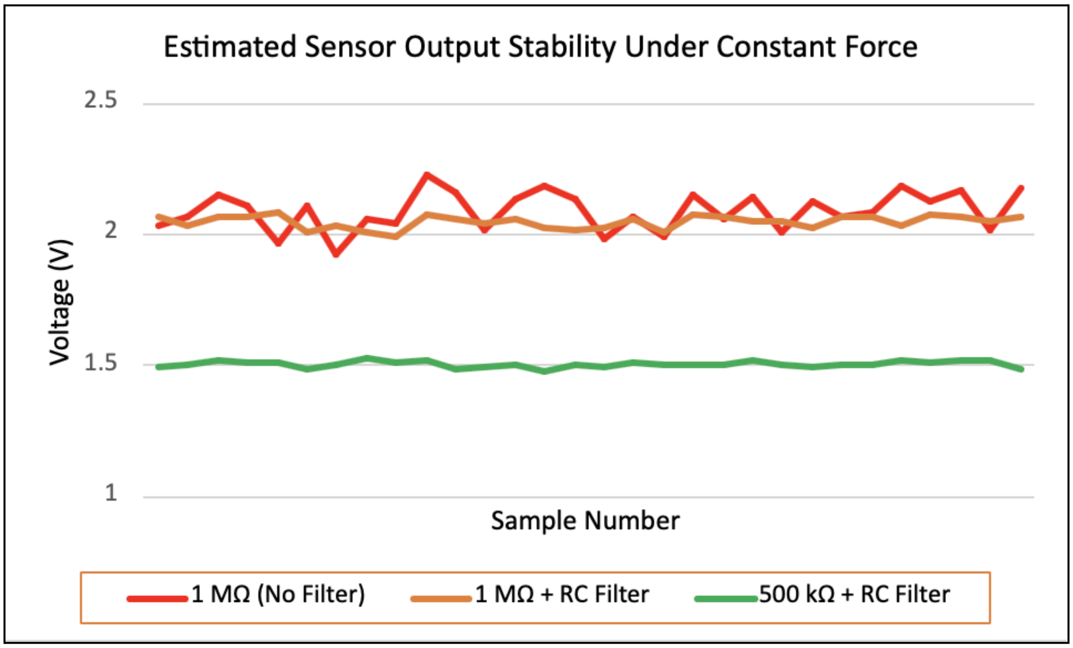

Figure 5. Estimated sensor output stability under constant-force conditions.

The 500 kΩ + RC filter configuration demonstrates the lowest output variation, confirming improved signal stability while maintaining adequate sensitivity.

Results and Reliability Impact: The initial implementation of RC filtering resulted in improved signal stability and reduced sensitivity to noise at the sensor output. The observed behavior confirms that high-impedance sensor nodes benefit significantly from simple filtering techniques, enhancing the reliability of analog measurements prior to wireless transmission.
This debugging process highlights several key engineering principles:

- High-impedance sensor interfaces require deliberate signal conditioning
- Reliability issues often emerge during hardware validation rather than initial design
- Targeted circuit-level improvements can significantly enhance real-world performance
- Iterative debugging is essential for developing robust embedded systems

IV. Design Iterations & Future Improvements

The initial prototype provided a functional proof of concept; however, several limitations were identified during early testing that impacted system performance and reliability. These findings guided subsequent design decisions and accelerated improvements in key areas of the system.

These included:

- Signal instability due to high-impedance sensor interfacing
- Unreliable connection interface
- Limited modularity in the programming architecture
- Increased sensitivity to noise in the analog signal path
- Challenges in maintaining consistent data acquisition under varying conditions

These limitations highlighted the need for improved signal conditioning, system modularity, and overall robustness to improve the product.

Implemented Improvements: Based on the limitations identified in the initial prototype, several targeted design improvements were implemented to enhance system reliability, signal integrity, and overall performance.

Key improvements included:

- **Implementation of RC Filtering:**
  A capacitor was added at the sensor output node to form a low-pass RC filter, reducing high-frequency noise and stabilizing the high-impedance signal prior to ADC sampling
- **Improved Sensor Interface Design (Mechanical & Electrical Stability):**
  The sensor contact pads were redesigned to improve mechanical consistency and electrical contact with the FSR sensors, resulting in more stable and repeatable measurements
- **Transition to ARM-Based Microcontroller:**
  The system was migrated from an ESP-based platform to an ARM-based microcontroller (nRF52840), providing improved peripheral control, enhanced ADC resolution and stability, and integrated BLE functionality for more reliable wireless communication
- **Modular System Architecture:**
  The design was updated to incorporate a modular architecture, eliminating the need to physically modify flex boards and enabling easier debugging, programming, and system integration through a detachable interface
- **Compact PCB Design:**
  The board layout was optimized to reduce overall size, resulting in a more compact form factor suitable for the intended flexible and embedded application

These improvements resulted in a more robust and reliable system, with enhanced signal stability, improved measurement consistency, and increased usability during development and deployment.

Future Improvements

While the current system demonstrates reliable performance, several opportunities exist for further enhancement:

- Integration of active signal conditioning (e.g., op-amp buffering) to reduce output impedance and improve measurement accuracy
- Implementation of sensor calibration techniques to establish a more precise relationship between applied force and measured output
- Additional shielding and grounding strategies to further mitigate noise in high-impedance analog paths
- Further miniaturization and integration of system components to improve wearability and usability
- Expansion of data processing and visualization capabilities for improved real-time analysis

These enhancements would further improve system accuracy, robustness, and overall usability in real-world applications. Overall, this project demonstrates the importance of iterative hardware design, signal conditioning, and system-level optimization in developing reliable embedded sensing systems.